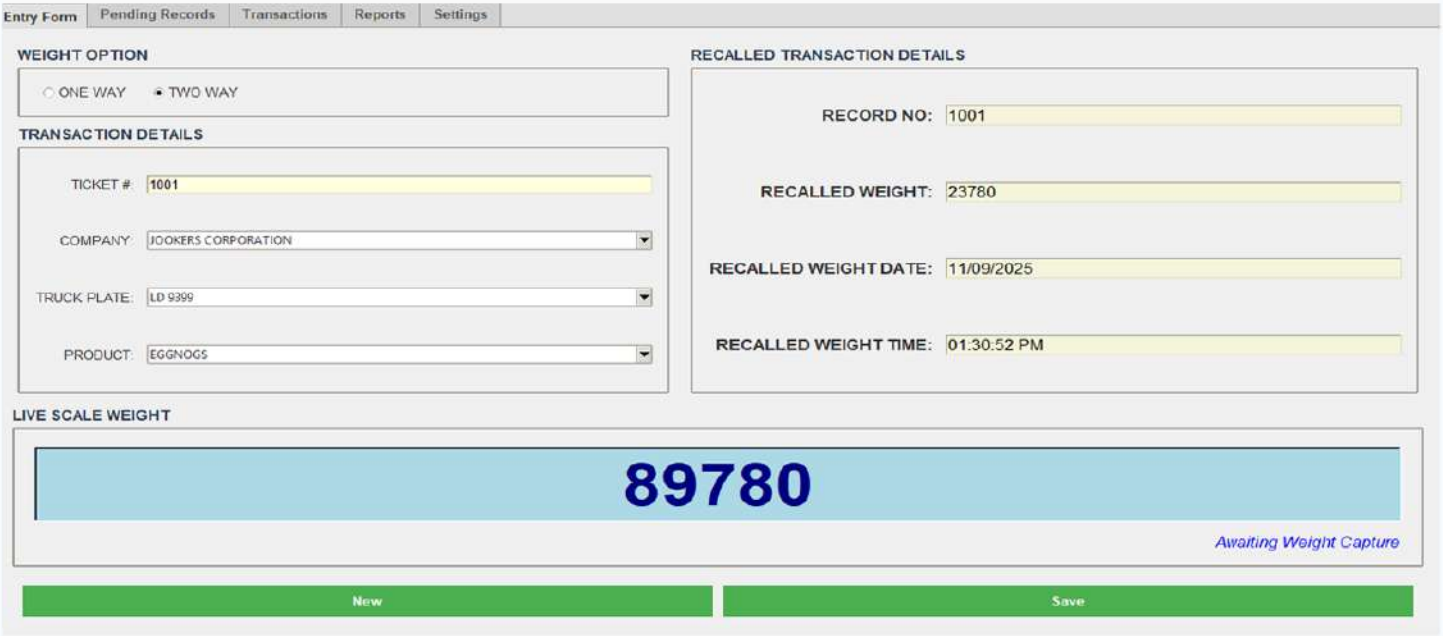
Truck Scale Weighing System

Operation Manual



A comprehensive guide for operators and administrators to install, configure, and operate the Truck Scale Weighing System safely and efficiently. This manual covers system overview, installation, configuration, daily operations, reporting, troubleshooting, and maintenance procedures.

Version 1.0 | 2025

Table of Contents

Truck Scale Weighing System - Operation Manual

1 System Overview 3

2 Key Features 4

3 Hardware Requirements 5

4 Software Installation 6

5 Initial Setup & Activation 7

6 Communication Ports Configuration 8

7 Master Data Management 9

8 Entry Form Operations 10

9 Weighing Modes 11

10 Customizing Entry Form 12

11 Pending Records Management 13

12 Transaction History 14

13 Reports & Export 15

14 Printing Receipts 16

15 User Authentication 17

16 Troubleshooting Guide 18

17 Error Messages & Resolution 19

18 Maintenance & Backup 20

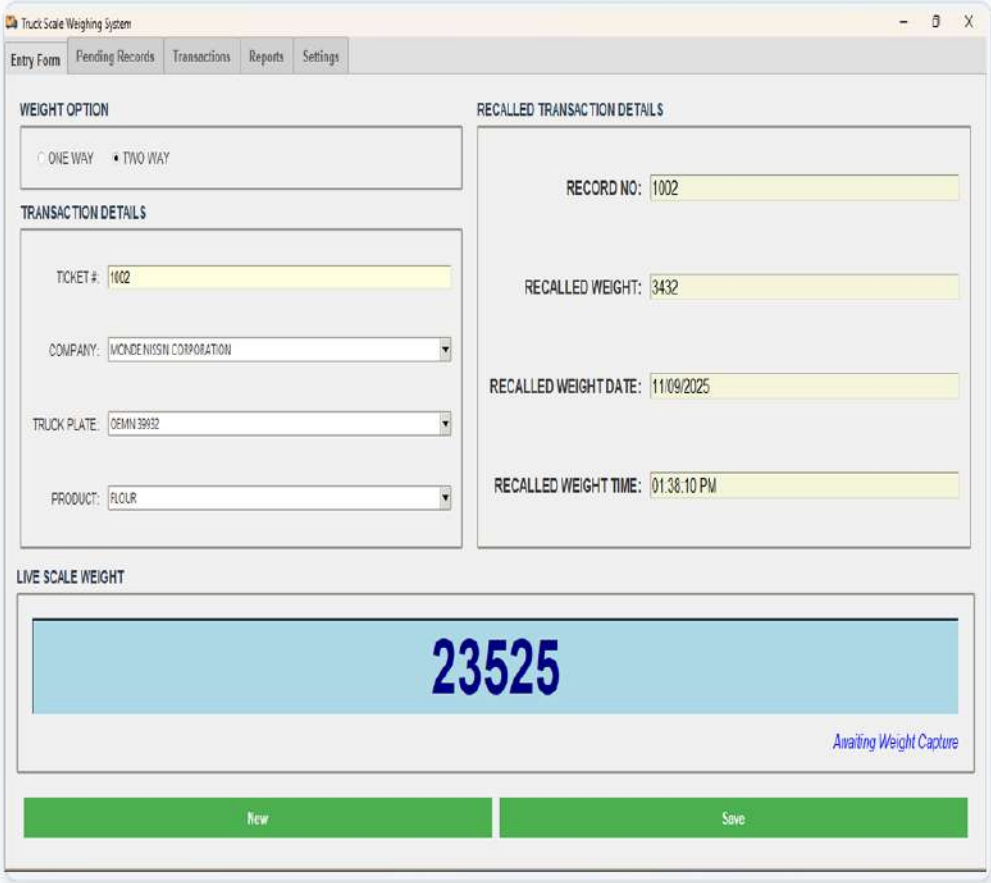
19 Support & Contact 20

System Overview

The application records truck weights from a serial-connected scale indicator, manages master data (companies, trucks, products, etc.), and produces tickets and reports.

Core components:

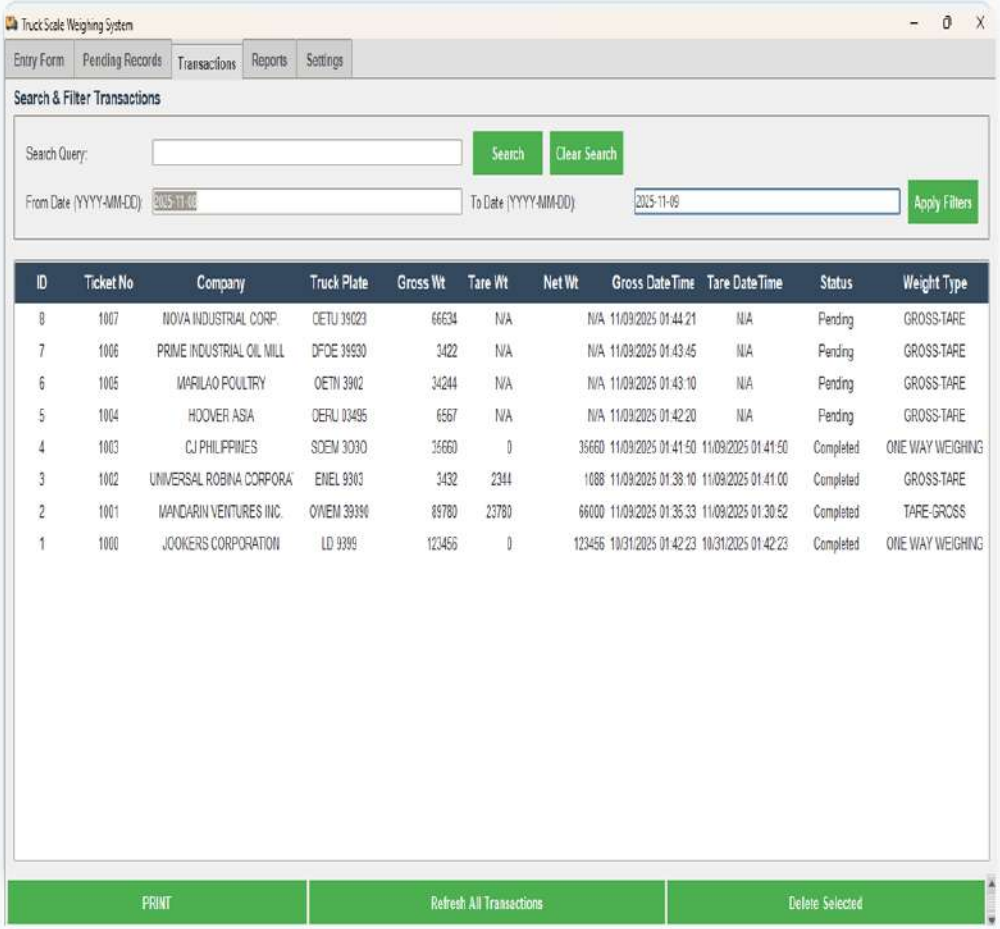
- Desktop UI (Tkinter)
- SQLite database storage
- Serial COM reader with regex parsing
- Optional Big Display output
- ReportLab printing to A6 receipts and CSV export



Key Features

The Truck Scale Weighing System offers comprehensive functionality for efficient weight measurement and record management:

- Live Scale reading with adjustable decimals and read interval
- One-way and Two-way (Gross/Tare/Net) workflows
- Pending records for second weighing recall
- Master Data lists for quick selection
- Transactions search/filter and CSV export
- Receipt printing with barcode support
- Admin activation and password change



ID	Ticket No	Company	Truck Plate	Gross Wt	Tare Wt	Net Wt	Gross DateTime	Tare DateTime	Status	Weight Type
8	1007	NOVA INDUSTRIAL CORP.	OETU 39023	66634	N/A	N/A	11/09/2025 01:44:21	N/A	Pending	GROSS-TARE
7	1006	PRIME INDUSTRIAL OIL MILL	DFOE 39930	3422	N/A	N/A	11/09/2025 01:43:45	N/A	Pending	GROSS-TARE
6	1005	MARILAO POULTRY	OETN 3902	34244	N/A	N/A	11/09/2025 01:43:10	N/A	Pending	GROSS-TARE
5	1004	HOOVER ASIA	OERU 03495	6567	N/A	N/A	11/09/2025 01:42:20	N/A	Pending	GROSS-TARE
4	1003	CJ PHILIPPINES	SOEM 3030	35660	0	35660	11/09/2025 01:41:50	11/09/2025 01:41:50	Completed	ONE WAY WEIGHING
3	1002	UNIVERSAL ROBINA CORPORA	ENEL 9303	3432	2344	1088	11/09/2025 01:38:10	11/09/2025 01:41:00	Completed	GROSS-TARE
2	1001	MANDARIN VENTURES INC.	OWEM 39390	89780	23780	66000	11/09/2025 01:35:33	11/09/2025 01:30:52	Completed	TARE-GROSS
1	1000	JOOKERS CORPORATION	LD 9399	123456	0	123456	10/31/2025 01:42:23	10/31/2025 01:42:23	Completed	ONE WAY WEIGHING

Hardware Requirements

The following hardware components are required for proper operation of the Truck Scale Weighing System:

- Windows 10/11 PC with administrator rights
- Serial COM port or USB-to-RS232 adapter for scale indicator
- Compatible scale indicator sending ASCII weight via serial
- Optional external big display (serial)
- Printer for A6/thermal receipts (USB/Network)
- Stable power and grounded connections

Important Serial Settings

For optimal communication with scale indicators, ensure your serial settings match the manufacturer's specifications:

- Baud Rate (typically 9600)
- Data Bits (typically 8)
- Parity (typically None)
- Stop Bits (typically 1)

Refer to your scale indicator manual for exact specifications.

Software Installation

1. **Download the Installation Package**

   Obtain the latest installation package from your vendor or the provided download link.

2. **Run the Installer**

   Right-click the installer and select "Run as administrator" to ensure proper installation.

3. **Accept Security Prompts**

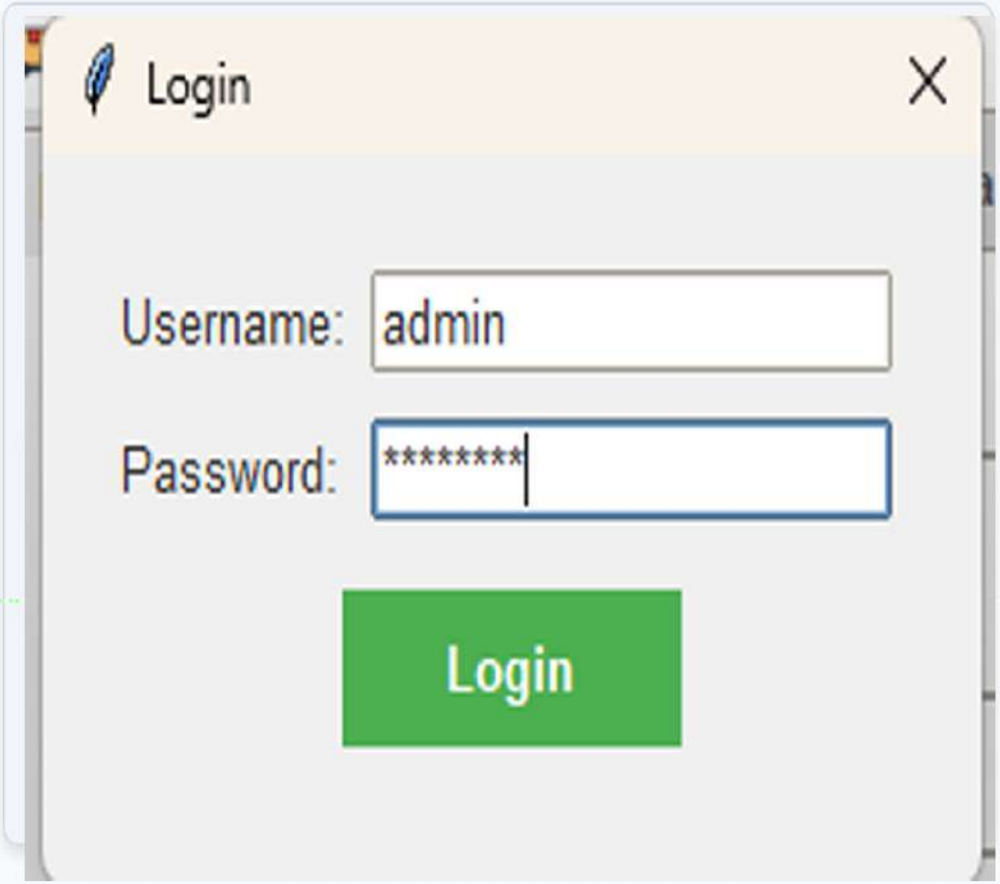
   When prompted, allow serial and printing permissions for full functionality.

4. **Complete Setup**

   Follow on-screen instructions to complete the installation process.

5. **Launch the Application**

   Start "Truck Scale Weighing System" from the desktop icon or start menu.



Note: Ensure printer drivers are installed and COM ports are identified in Device Manager before configuring the application.

Initial Setup & Activation

1 Start the Application

Launch the Truck Scale Weighing System and log in with the administrator credentials provided by your IT department or installer.

2 Access Activation Tab

Navigate to Settings > Activation to view the current activation status and manage your license.

3 Enter Activation Code

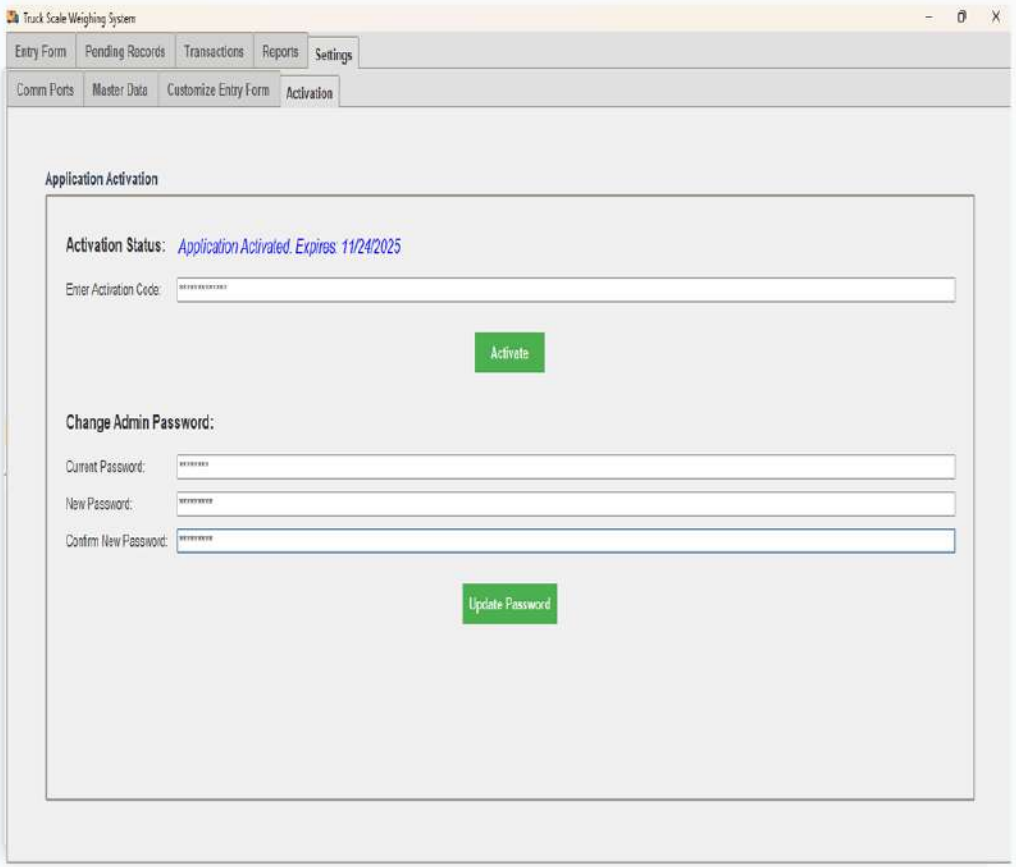
Type or paste your provided activation code in the field and click Activate. Verify that the expiration date appears.

4 Change Admin Password

Enter your current password, create a new secure password, confirm it, and click Update Password to enhance security.

5 Record Expiry Information

Note the expiration date displayed after activation and maintain a record for future renewal planning.



Important: Store your activation code in a secure location. Contact your vendor before the expiration date to obtain renewal instructions.

Communication Ports Configuration

1 Open Settings > Comm Ports

Access the Communication Ports configuration panel from the main menu.

2 Main Scale Connection

Configure the connection to your primary weighing scale:

- Select COM Port and Baud Rate from dropdown lists
- Set Data Format Regex (default provided) and decimal places
- Click Auto-Detect if unsure of settings
- Click Start Connection to connect to the scale

3 Big Display Output (Optional)

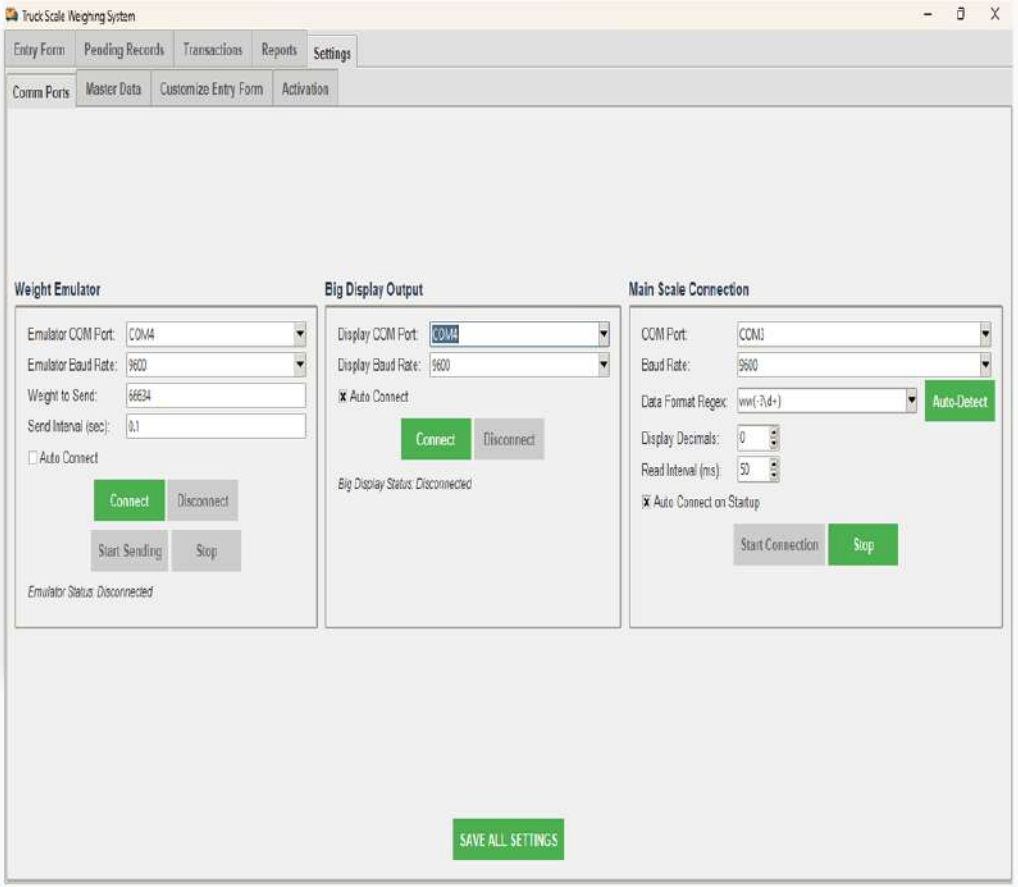
If using an external display panel:

- Select Display COM Port and Baud Rate
- Check Auto Connect if desired
- Click Connect to establish communication

4 Weight Emulator (Testing Only)

For testing without a physical scale:

- Configure Emulator COM Port and Baud Rate
- Set Weight to Send and Send Interval
- Connect and Start Sending for simulation



Note: Always verify the correct COM ports in Device Manager before configuring. Use the "Status:" indicators to confirm successful connections.

Master Data Management

1 Access Master Data

Navigate to Settings > Master Data to access all data lists that will be available in the Entry Form dropdown menus.

2 Select Data Category

Choose which list to manage: Companies, Trucks, Products, Drivers, Origins, Destinations, Designations, or Senders.

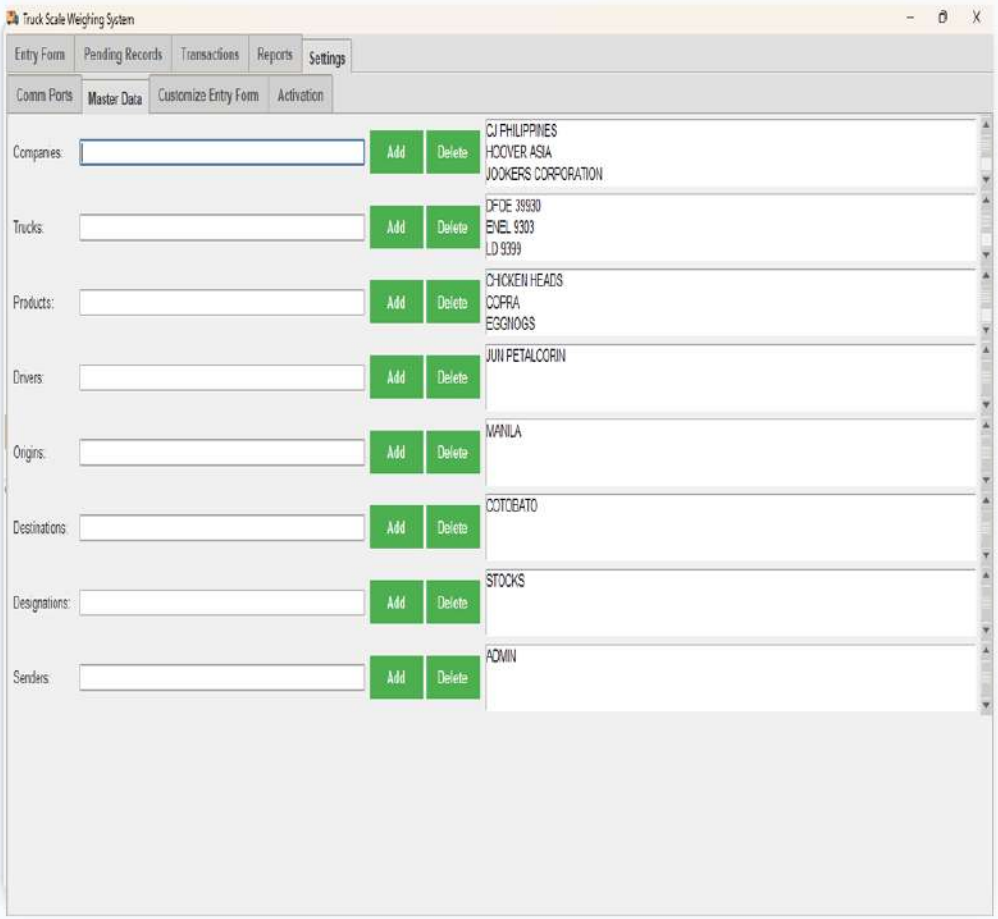
3 Add New Items

Type the new value in the input field and click the Add button to include it in the selected list.

4 Delete Existing Items

Select an item from the list on the right and click the Delete button to remove it from the system.

> **Tip:** Keep naming conventions consistent and avoid duplicates to maintain data integrity. Changes to master data are immediately available in Entry Form dropdowns.



Entry Form Operations

1 Access the Entry Form

Navigate to the Entry Form tab from the top menu navigation bar.

2 Select Weighing Mode

Choose between One Way or Two Way weighing options based on your operational requirements.

3 Complete Transaction Details

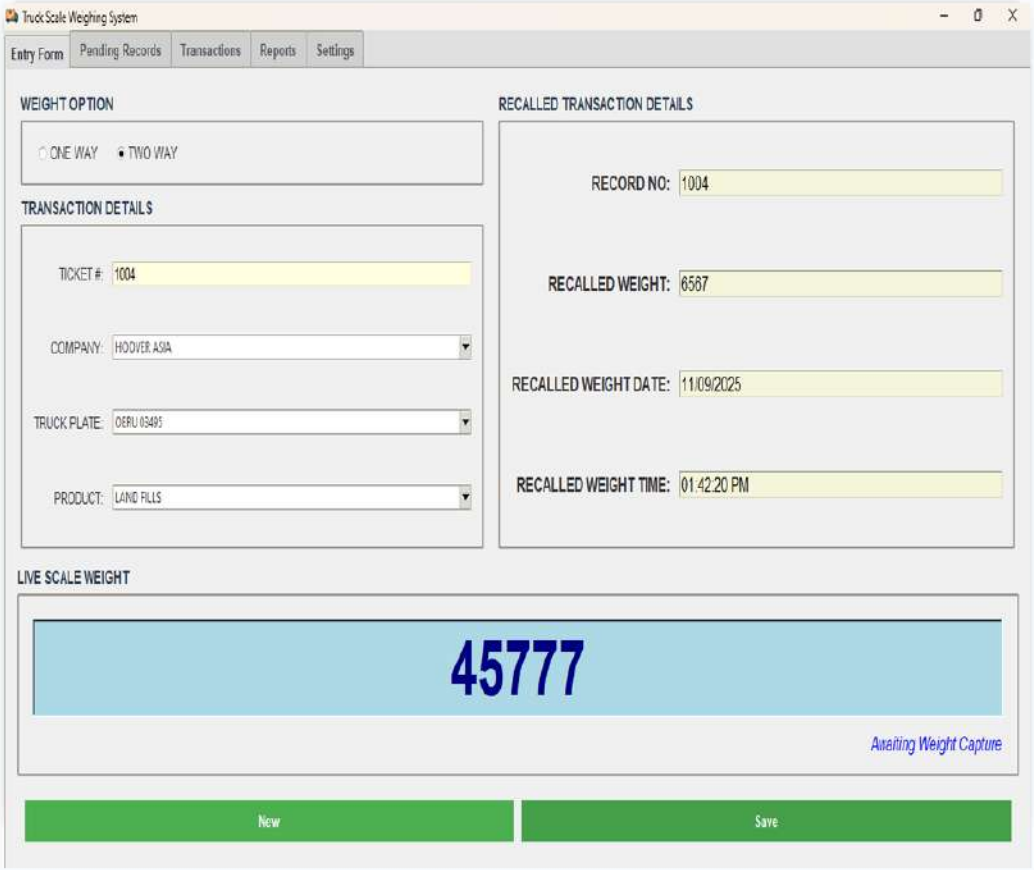
Fill in required fields such as Company, Truck Plate, Product, and other details using the dropdown selections.

4 Verify Live Scale Weight

Ensure the truck is properly positioned on the scale and the displayed weight reading has stabilized.

5 Save the Transaction

Click the Save button to record the weighing. Use New button to clear the form and start another transaction.



Note: The Recalled Transaction Details panel will display information when recalling pending records for second weighing operations.

Weighing Modes: One-way vs Two-way

One-way Weighing

Single pass weighing process

- Used for single pass weighing operations (inbound gross only)
- Records a completed transaction with a single weight measurement
- Simplified workflow with immediate completion
- Useful for inventory counts and single-measurement applications

Two-way Weighing

Gross/Tare/Net calculation process

- First pass saves as Pending (Gross or Tare) with assigned ticket number
- Second pass recalls the pending record and calculates Net weight
- Allows accurate weight difference calculation between arrival and departure
- Ideal for loading/unloading operations requiring accurate net weight

VS

Best Practice:

Train operators to verify ticket numbers and truck plates before finalizing second weighing to ensure accurate pairing of weight measurements.

Customizing the Entry Form

1 Access Customization Settings

Navigate to Settings > Customize Entry Form in the main menu.

2 Select Fields to Display

Use the Add/Remove buttons to move fields between Available Fields and Selected Fields (Entry Order) lists.

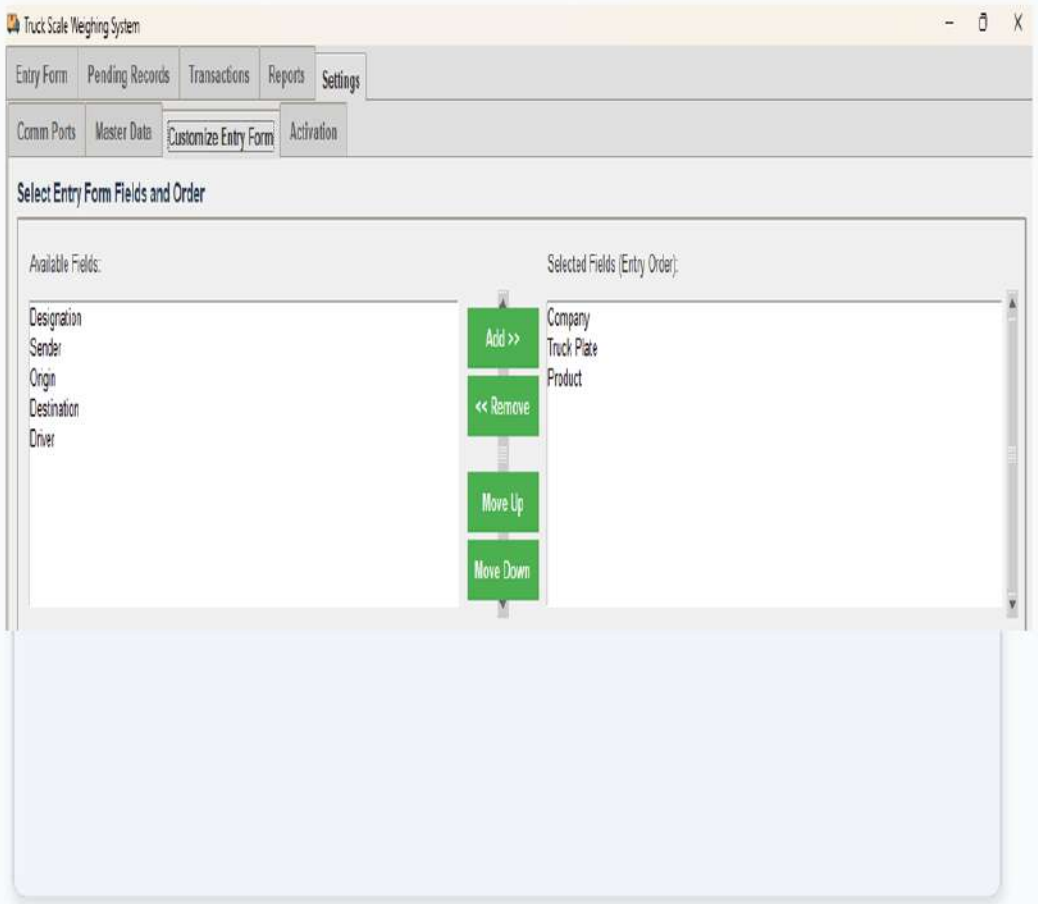
3 Arrange Field Order

Select a field in the Selected Fields list and use Move Up/Move Down buttons to arrange the display order on the Entry Form.

4 Save Your Configuration

Click the SAVE ENTRY FORM LAYOUT button to apply your changes.

> **Note:** Only fields in the Selected Fields list will appear on the Entry Form. If you need to revert to the default configuration, click Reset Entry Form to Default.



Managing Pending Records

1 Open Pending Records

Navigate to the Pending Records tab from the main menu to view all incomplete weighing transactions.

2 Review the List

Examine records by Ticket No, Company, Truck Plate, and Gross Date/Time to identify the required transaction.

3 Complete Second Weighing

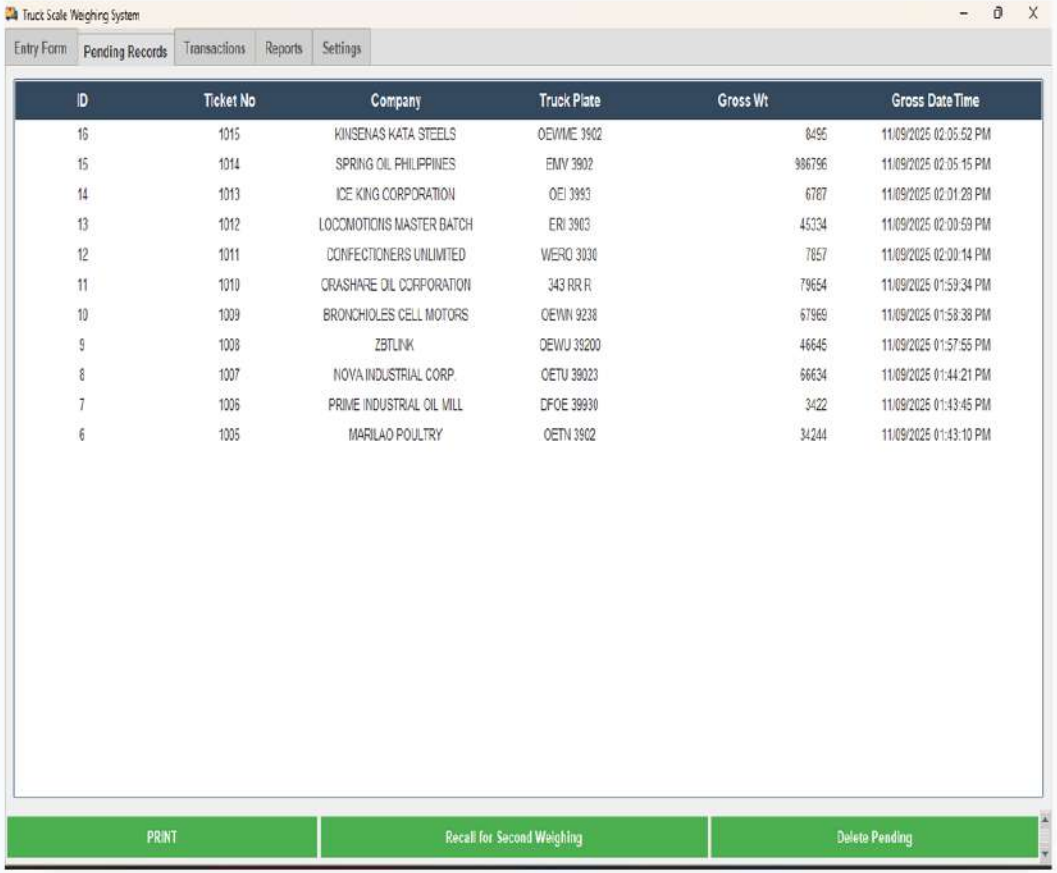
Select a record and click "Recall for Second Weighing" to load it into the Entry Form. Complete the transaction and Save.

4 Remove Errors

To delete incorrect records, select them from the list and click "Delete Pending".

5 Print Records

Use the "PRINT" button to generate a physical copy of the pending records list when needed.



ID	Ticket No	Company	Truck Plate	Gross Wt	Gross Date Time
16	1015	KINSENAS KATA STEELS	OEWME 3902	8495	11/09/2025 02:05:52 PM
15	1014	SPRING OIL PHILPPINES	EMV 3902	986796	11/09/2025 02:05:15 PM
14	1013	ICE KING CORPORATION	OEI 3993	6787	11/09/2025 02:01:28 PM
13	1012	LOCOMOTIONS MASTER BATCH	ERI 3903	45334	11/09/2025 02:00:59 PM
12	1011	CONFECTIONERS UNLIMITED	WERO 3030	7857	11/09/2025 02:00:14 PM
11	1010	CRASHARE OIL CORPORATION	343 RR R	79654	11/09/2025 01:59:34 PM
10	1009	BRONCHIOLES CELL MOTORS	OEWN 9238	67969	11/09/2025 01:58:38 PM
9	1008	ZBTLINK	OEWU 39200	46645	11/09/2025 01:57:55 PM
8	1007	NOVA INDUSTRIAL CORP.	OETU 39023	66634	11/09/2025 01:44:21 PM
7	1006	PRIME INDUSTRIAL OIL MILL	DFOE 39930	3422	11/09/2025 01:43:45 PM
6	1005	MARILAO POULTRY	OETN 3902	34244	11/09/2025 01:43:10 PM

Tip: Two-way weighing transactions remain in Pending Records until the second weighing is complete. Regularly check for outdated records that may need attention.

Transaction History

1 Navigate to Transactions Tab

Click on the "Transactions" tab in the main navigation menu to view the complete transaction history.

2 Search and Filter Transactions

Use the Search Query field to filter by company name, truck plate, product, or ticket number for quick access to specific records.

3 Apply Date Range

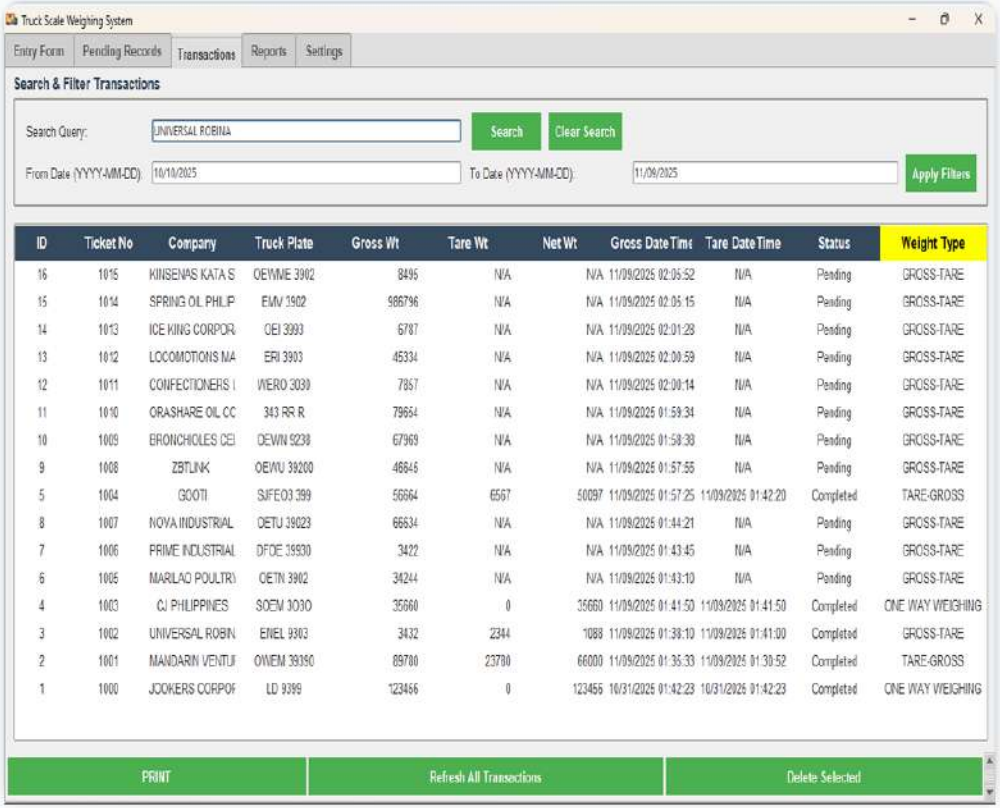
Set From Date and To Date fields to narrow results to a specific time period, then click "Apply Filters" to update the view.

4 Review Transaction Details

View important information including ID, Ticket No, Company, Truck Plate, weights, dates/times, and Status (Pending or Completed).

5 Perform Actions

Use action buttons at the bottom to Print selected transactions, Refresh All Transactions, or Delete Selected records as needed.



ID	Ticket No	Company	Truck Plate	Gross Wt	Tare Wt	Net Wt	Gross Date Time	Tare Date Time	Status	Weight Type
16	1015	KINSENAS KATA S	OEYMNE 3902	8495	N/A	N/A	11/09/2025 02:06:52	N/A	Pending	GROSS-TARE
15	1014	SPRING OIL PHILIP	EMV 3902	986796	N/A	N/A	11/09/2025 02:06:15	N/A	Pending	GROSS-TARE
14	1013	ICE KING CORPOR	OEI 3993	6787	N/A	N/A	11/09/2025 02:01:28	N/A	Pending	GROSS-TARE
13	1012	LOCOMOTIONS MA	ERI 3903	45334	N/A	N/A	11/09/2025 02:00:59	N/A	Pending	GROSS-TARE
12	1011	CONFECTIONERS I	WERO 3030	7857	N/A	N/A	11/09/2025 02:00:14	N/A	Pending	GROSS-TARE
11	1010	ORASHARE OIL CC	343 RR R	79654	N/A	N/A	11/09/2025 01:59:34	N/A	Pending	GROSS-TARE
10	1009	BRONCHIOLES CEI	OEWN 9238	67969	N/A	N/A	11/09/2025 01:58:38	N/A	Pending	GROSS-TARE
9	1008	ZBTLINK	OEWU 39200	46645	N/A	N/A	11/09/2025 01:57:55	N/A	Pending	GROSS-TARE
5	1004	GOOTI	SIFEO3 399	56664	6567	50097	11/09/2025 01:57:25	11/09/2025 01:42:20	Completed	TARE-GROSS
8	1007	NOVA INDUSTRIAL	OETU 39023	66634	N/A	N/A	11/09/2025 01:44:21	N/A	Pending	GROSS-TARE
7	1006	PRIME INDUSTRIAL	DFOE 39930	3422	N/A	N/A	11/09/2025 01:43:45	N/A	Pending	GROSS-TARE
6	1005	MARILAO POULTRY	OETN 3902	34244	N/A	N/A	11/09/2025 01:43:10	N/A	Pending	GROSS-TARE
4	1003	CJ PHILIPPINES	SOEM 3030	35660	0	35660	11/09/2025 01:41:50	11/09/2025 01:41:50	Completed	ONE WAY WEIGHING
3	1002	UNIVERSAL ROBIN	EMEL 9303	3432	2344	1088	11/09/2025 01:38:10	11/09/2025 01:41:00	Completed	GROSS-TARE
2	1001	MANDARIN VENTUI	OWEM 39390	89780	23780	66000	11/09/2025 01:36:33	11/09/2025 01:30:52	Completed	TARE-GROSS
1	1000	JOOKERS CORPOF	LD 9399	123466	0	123466	10/31/2025 01:42:23	10/31/2025 01:42:23	Completed	ONE WAY WEIGHING

Note: The Status column shows whether a transaction is "Pending" (awaiting second weighing) or "Completed" (has all required weight data).

Reports & CSV Export

1 Open Reports Tab

Navigate to the Reports tab in the main menu to view transaction history in a structured format.

2 Apply Filters

Use the Search Query field to filter by company, plate, product, or ticket. Set a date range with From Date and To Date fields.

3 Export Filtered Data

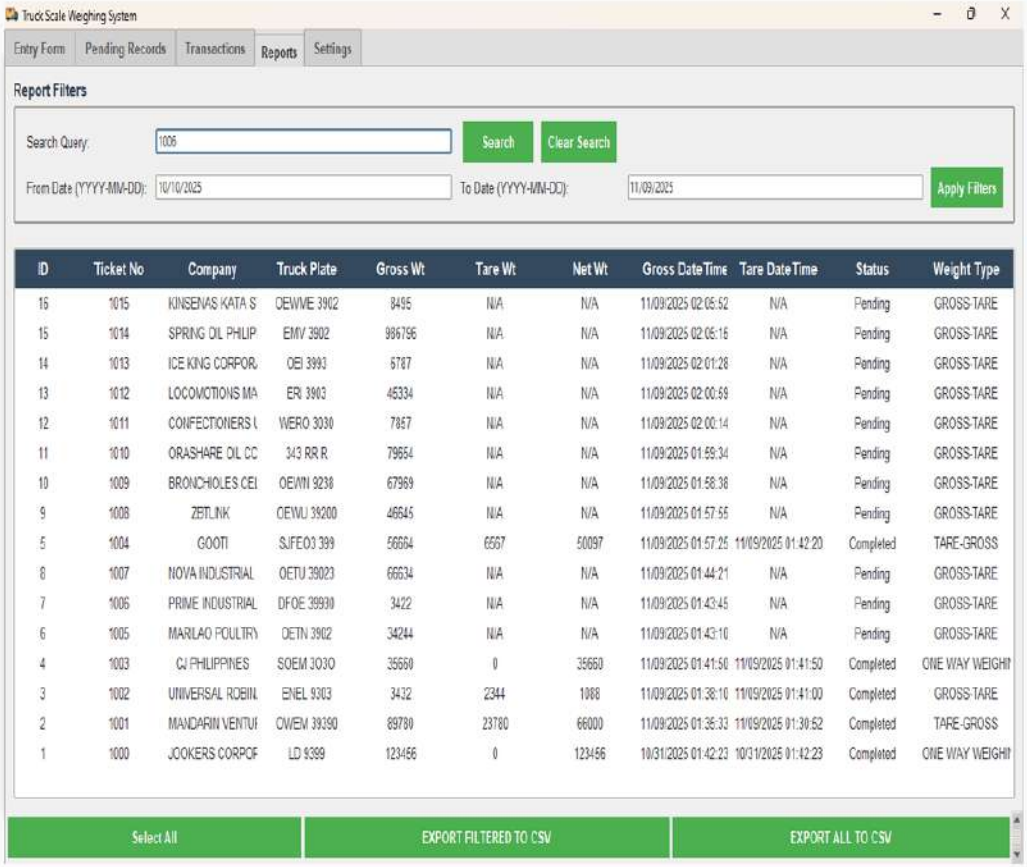
Click "EXPORT FILTERED TO CSV" to save only the currently displayed records based on your search criteria.

4 Export Complete Dataset

Use "EXPORT ALL TO CSV" to create a comprehensive file with all transaction records regardless of current filters.

5 Use Batch Selection

Click "Select All" when you need to perform operations on multiple records simultaneously.



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16	1015	KINSENAS KATA S	OEWME 3902	8495	N/A	N/A	11/09/2025 02:05:52	N/A	Pending	GROSS-TARE
15	1014	SPRING OIL PHILIP	EMV 3902	986796	N/A	N/A	11/09/2025 02:05:16	N/A	Pending	GROSS-TARE
14	1013	ICE KING CORPORA	OEI 3993	6787	N/A	N/A	11/09/2025 02:01:28	N/A	Pending	GROSS-TARE
13	1012	LOCOMOTIONS MA	ERI 3903	45334	N/A	N/A	11/09/2025 02:00:59	N/A	Pending	GROSS-TARE
12	1011	CONFECTIONERS I	WERO 3030	7857	N/A	N/A	11/09/2025 02:00:14	N/A	Pending	GROSS-TARE
11	1010	ORASHARE OIL CC	343 RR R	79654	N/A	N/A	11/09/2025 01:59:34	N/A	Pending	GROSS-TARE
10	1009	BRONCHIOLES CEI	OEWN 9238	67969	N/A	N/A	11/09/2025 01:58:38	N/A	Pending	GROSS-TARE
9	1008	ZEITLINK	OEWU 39200	46645	N/A	N/A	11/09/2025 01:57:55	N/A	Pending	GROSS-TARE
5	1004	GOOTI	SJFEO3 399	56664	6567	50097	11/09/2025 01:57:25	11/09/2025 01:42:20	Completed	TARE-GROSS
8	1007	NOVA INDUSTRIAL	OETU 39023	66634	N/A	N/A	11/09/2025 01:44:21	N/A	Pending	GROSS-TARE
7	1006	PRIME INDUSTRIAL	DFOE 39930	3422	N/A	N/A	11/09/2025 01:43:45	N/A	Pending	GROSS-TARE
6	1005	MARILAO POULTRY	OETN 3902	34244	N/A	N/A	11/09/2025 01:43:10	N/A	Pending	GROSS-TARE
4	1003	CJ PHILIPPINES	SOEM 3030	35660	0	35660	11/09/2025 01:41:50	11/09/2025 01:41:50	Completed	ONE WAY WEIGHT
3	1002	UNIVERSAL ROBIN	ENEL 9303	3432	2344	1088	11/09/2025 01:38:10	11/09/2025 01:41:00	Completed	GROSS-TARE
2	1001	MANDARIN VENTUI	OWEM 39390	89780	23780	66000	11/09/2025 01:36:33	11/09/2025 01:30:52	Completed	TARE-GROSS
1	1000	JOOKERS CORPOF	LD 9399	123456	0	123456	10/31/2025 01:42:23	10/31/2025 01:42:23	Completed	ONE WAY WEIGHT

Tip: When exporting to CSV, keep a read-only master copy for auditing purposes. This prevents accidental changes to your historical data when opening in Excel or other spreadsheet applications.

Printing Receipts

Where to print:

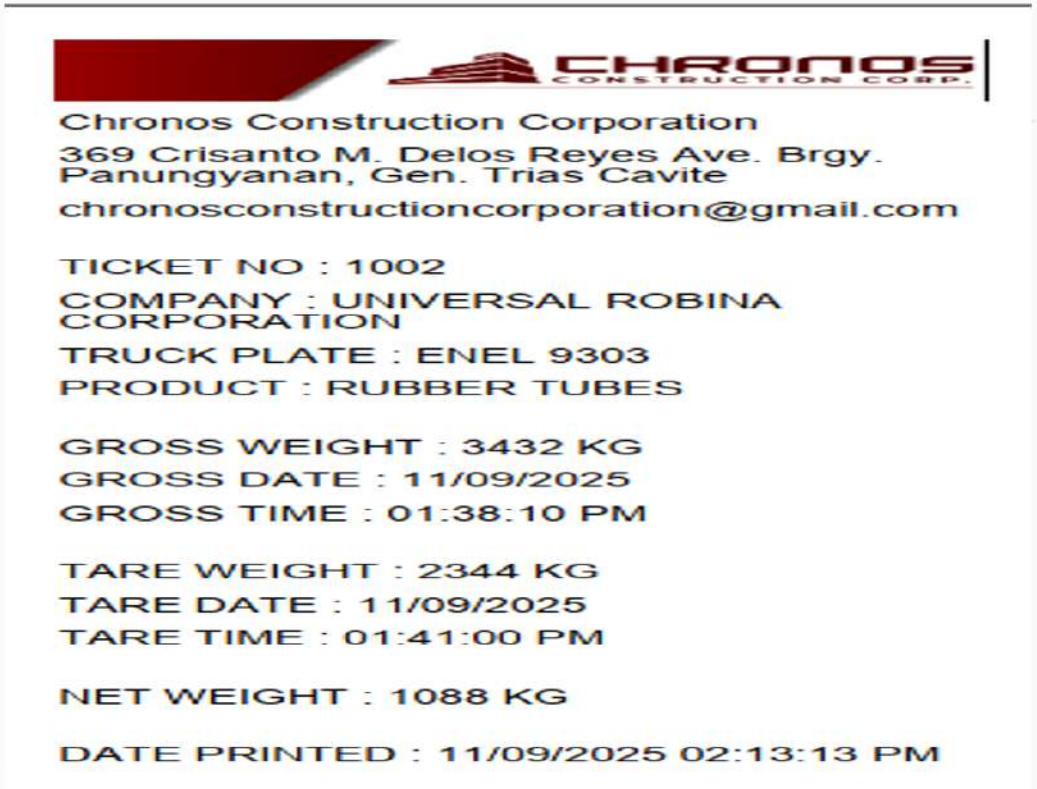
Access printing functionality from Transactions and Pending Records screens via the PRINT button at the bottom of these interfaces.

What's on the receipt:

- Ticket number and date/time stamps
- Company name and truck plate
- Product and weighing details
- Gross, tare and net weights

Printing steps:

1. Select the target record(s) from the list
2. Click PRINT and select printer from dialog
3. Verify paper size (A6 or thermal) and alignment



Chronos Construction Corporation
369 Crisanto M. Delos Reyes Ave. Brgy. Panungyanan, Gen. Trias Cavite
chronosconstructioncorporation@gmail.com

TICKET NO : 1002
COMPANY : UNIVERSAL ROBINA CORPORATION
TRUCK PLATE : ENEL 9303
PRODUCT : RUBBER TUBES

GROSS WEIGHT : 3432 KG
GROSS DATE : 11/09/2025
GROSS TIME : 01:38:10 PM

TARE WEIGHT : 2344 KG
TARE DATE : 11/09/2025
TARE TIME : 01:41:00 PM

NET WEIGHT : 1088 KG

DATE PRINTED : 11/09/2025 02:13:13 PM

⚠ Troubleshooting: If printing fails, verify printer drivers are installed and not paused. Test with a Windows test page to isolate software vs hardware issues.

User Authentication

Login

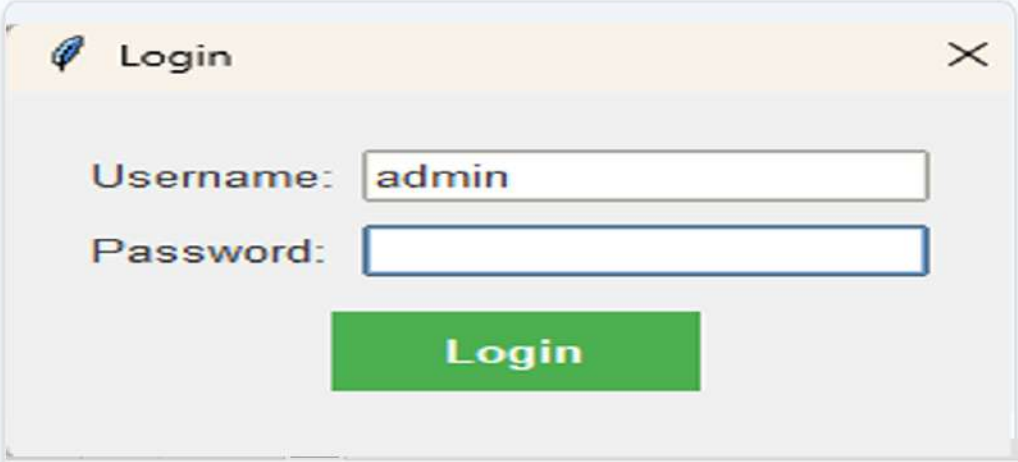
- On launch, enter Username and Password and click Login
- Default admin credentials are provided by your system installer

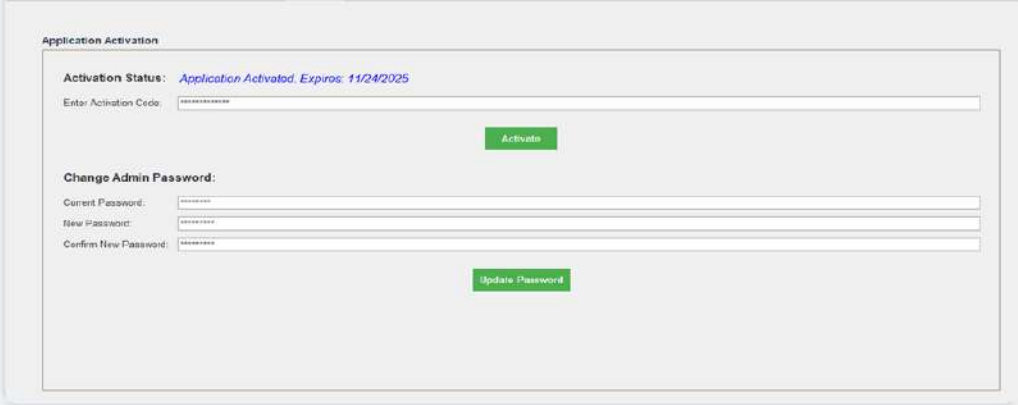
Admin Tasks

- Change password under Settings > Activation (Change Admin Password section)
- Manage license activation and verify system expiry date

Best Practices

- Use strong passwords with mixed characters
- Rotate passwords periodically (every 90 days)
- Restrict admin access to authorized personnel only





Troubleshooting Guide

No live weight

- Check COM Port, Baud Rate, and cable; verify with Device Manager.
- Confirm Data Format Regex matches indicator output and decimals setting.
- Use Emulator to validate app workflow.

Unstable/incorrect values

- Increase Read Interval and verify indicator's stable flag behavior.

Printing fails

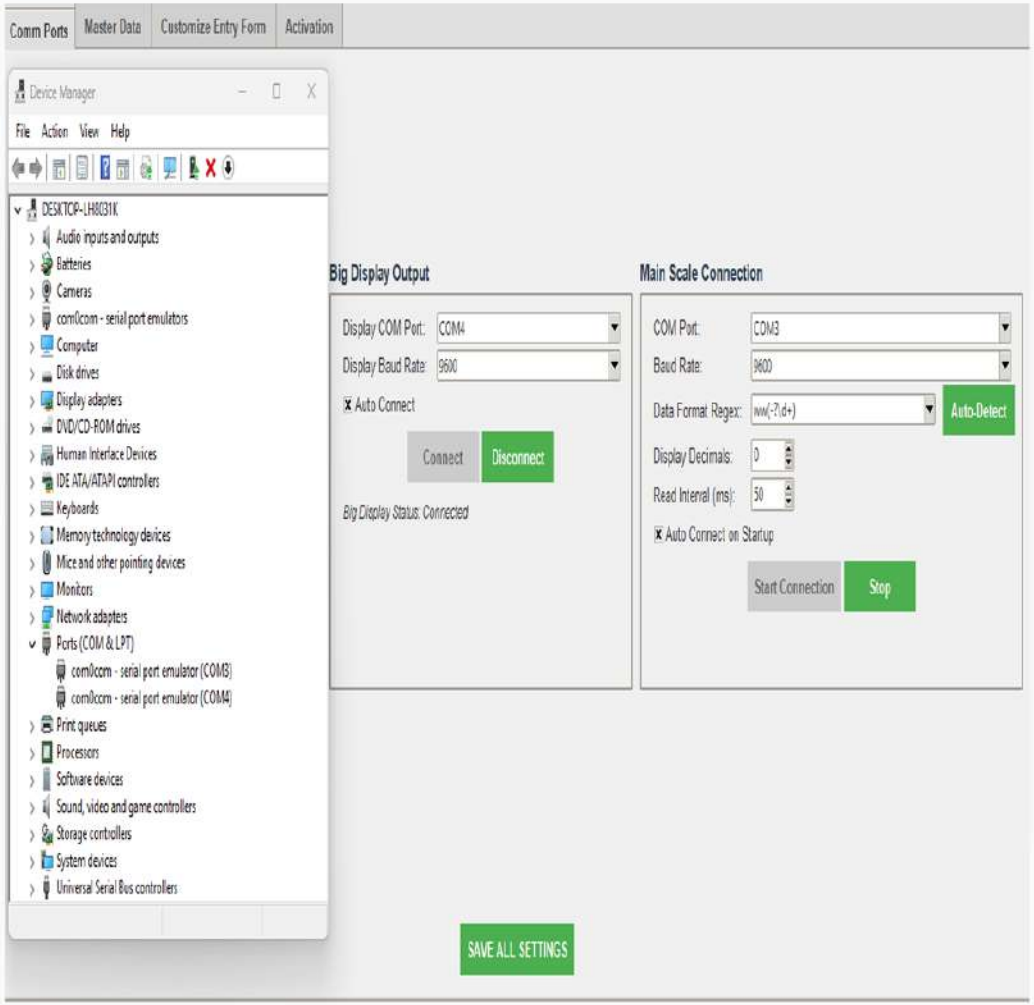
- Reinstall printer drivers; test a Windows test page.

CSV won't open correctly

- Open with UTF-8 and set delimiter to comma.

Database is slow/locked

- Close other instances; avoid network shares.



Error Messages & Resolution

Typical messages and actions:

Error Message	Resolution
Cannot open COM port	Port in use or wrong; close other apps and reselect.
Invalid activation code / expired license	Re-enter correct code or contact support.
Required fields missing	Complete mandatory Entry Form fields.
No record selected	Select a row before printing or deleting.
Export failed	Verify write permissions to destination folder.
Database error	Ensure disk space and folder permissions; retry after restarting the app.

Most error messages provide information about which operation failed and why. Always check connection status and permissions when troubleshooting.

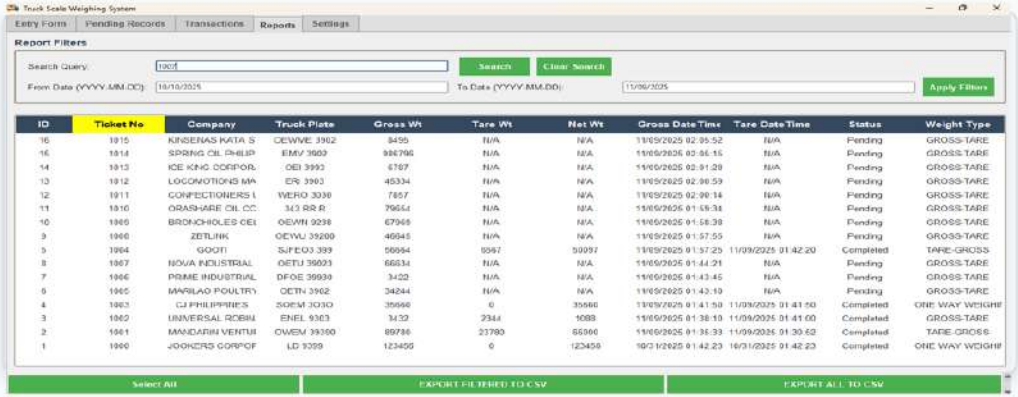
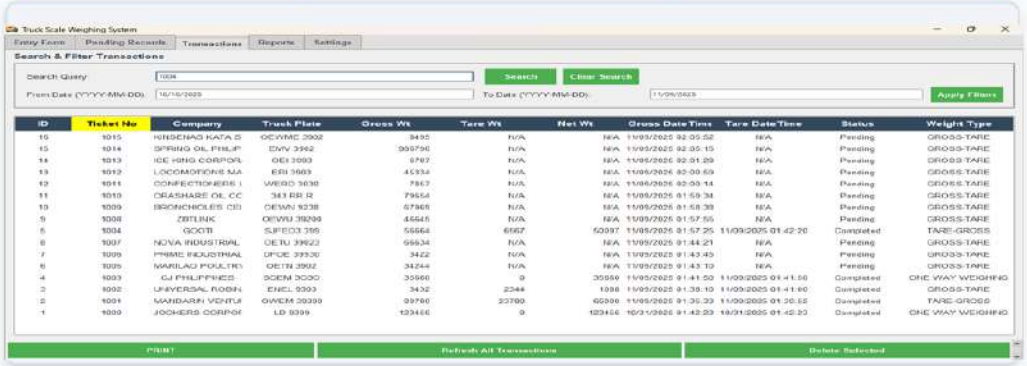
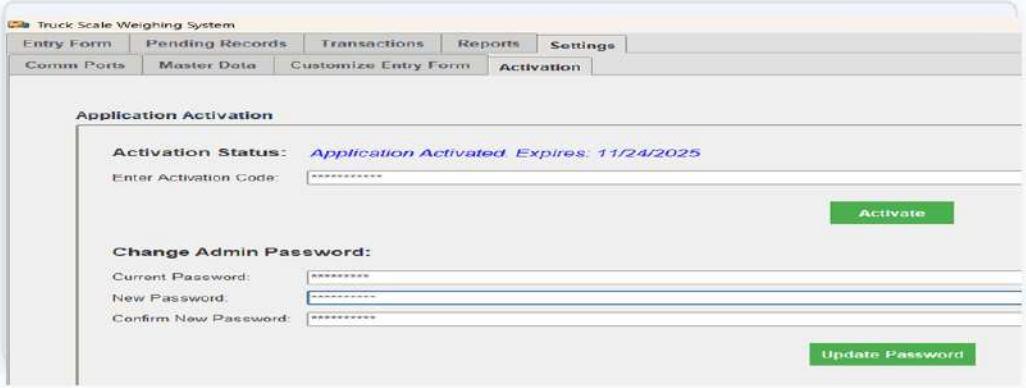
Maintenance, Backup, and Support

Maintenance & Backup

- Database: Regularly copy the SQLite database file from the application data folder to a secure backup location while the app is closed.
- Periodic CSV exports for archival.
- Keep Windows, USB-Serial drivers, and printers updated.
- Clean serial connectors; verify grounding.

Logs & Updates

- Keep a log of ticket ranges and activations; update the app per vendor guidance.

Support & Contact

- For license renewal, hardware compatibility, or technical issues, contact your system provider or IT helpdesk. Provide: app version, Windows version, COM settings, and screenshots of the issue.